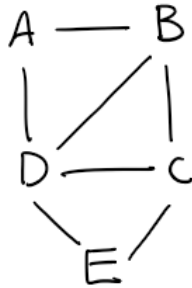


Lesson-5: Quiz (graded)

1

Multiple choice 1 point

What is the transitivity of the network below?



- ☐ 9/14
- ☐ 5/8
- ☐ 11/15
- ☐ 23/30

2

Fill in the Blank 1 point

Select the right choice for following statements regarding the CPL and efficiency of an undirected and connected network.

1. The average shortest-path length is less than or equal to the diameter

2. As the CPL increases, the related efficiency decreases

3. The efficiency of a complete graph with 10 nodes is 1

4. As the CPL increases, the diameter increases.

☐ False☐ True☐ False☐ True☐ False☐ True☐ False☐ True

3

Multiple choice 1 point

Which of the following statements about the Watts-Strogatz model is True?

- ☐ As the rewiring probability p increases, the CPL decreases linearly
- ☐ In the WS model, to have strong clustering (similar to a lattice network with the same average degree), the rewiring probability should be more than about 10%
- ☐ As the rewiring probability p increases, the diameter decreases logarithmically
- ☐ If the rewiring probability of the WS model is sufficiently large, the network diameter will be at the same level with the corresponding $G(n,p)$ network

4

Multiple choice 1 point

Which of the following statements about the Preferential Attachment (PA) is True?

- ☐ The CPL of the PA model (degree exponent = 3) has the same asymptotic growth with the CPL of $G(n,p)$
- ☐ The diameter of the PA model increases slower than the diameter of $G(n,p)$
- ☐ The average clustering coefficient of the PA model decreases faster compared to the $G(n,p)$ model
- ☐ The average clustering coefficient of the PA model is the same with the $G(n,p)$ model

5

Multiple choice 1 point

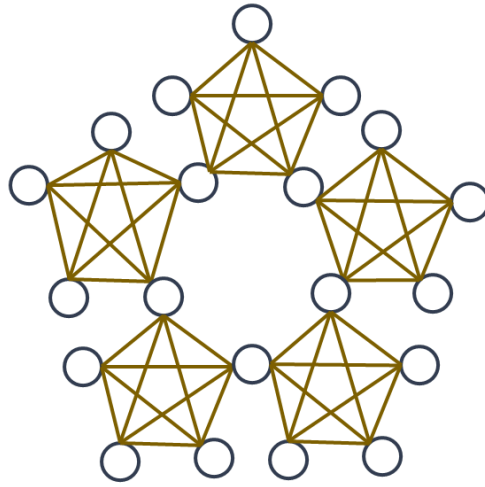
Which of the following statements about network motifs is True?

- ☐ The directed configuration model can generate random directed networks preserving the in-degree and out-degree of each node
- ☐ The method we discussed in the lecture uses random sampling without replacement to compute statistically significant motifs
- ☐ Suppose that $m(G)$ is the number of “mutual out” instances in the given graph G , and $m(G_r)$ is the random variable that counts the number of “mutual out” instances in a random graph. If the probability that $m(G_r)$ is larger than $m(G)$ is 99%, the “mutual in” motif is more common in G than expected by chance with a 99% statistical confidence

6

Multiple choice 1 point

First, calculate the CPL and the diameter for the following network. How large is the difference: diameter minus CPL? Round to 2 decimal places.



- ☐ 0.95
- ☐ 0.70
- ☐ 1.05
- ☐ 0.85